

Youssef M. Hussein

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EDUCATION

Ph.D. in Computer Science | University of Minnesota, Twin Cities | September 2023 – Present | Minneapolis, MN

- Research focus: efficient native-support systems for spatiotemporal data and enabling large language models to better understand and interact with such data.
- GAGE Fellowship, Grants to Advance Graduate Education, 2023–2024.

M.S. in Computer Science | University of Minnesota, Twin Cities | September 2023 – April 2025 | Minneapolis, MN

- Relevant Coursework: Advanced Database Systems, Operating Systems, Advanced Spatial Data Science.

B.Sc. in Computer Engineering | The American University in Cairo | September 2018 – June 2023 | Cairo, Egypt

- Senior Thesis: GPU Accelerated Dataflow Analysis.
- Dean's Honorary List: Spring 2021, Fall 2021, Spring 2022, Fall 2022, Spring 2023.
- Al-Ghurair Foundation Full Merit-Based Scholarship, 2019–2023.

RESEARCH EXPERIENCE

Graduate Research Assistant & Fellow | University of Minnesota, Twin Cities | September 2023 – Present | Minneapolis, MN

- Conduct research on efficient data systems with native support for spatiotemporal data and on enabling large language models to better understand and interact with such data.
- KAFY: Developed an extensible and scalable transformers-based system for trajectory data analysis, supporting trajectory summarization, generation, prediction, imputation, and classification.
- LLMs for Spatial Analysis: Presented several tutorials researching large language models for spatial analysis queries.
- POLARIS/iHARP: Developed infrastructure and web-based interfaces for interactive querying and visualization of scientific raster data.

Research Intern — GeoAI Team | Oak Ridge National Laboratory (ORNL) | April 2026 – August 2026 | Oak Ridge, TN

- Conducted research with the GeoAI team on AI and geospatial data applications.
- Contributed to research on multimodal Earth observation data and agentic reasoning systems.
- Research contributions resulted in publications at NeurIPS and ACM SIGSPATIAL.
- Aether: Developed an agentic retrieval and reasoning system for multimodal Earth observation data.
- ARFA: Developed an agentic system for real-time riverine flood response using spatial data and geospatial AI agents.

Research Engineer & Junior Research Assistant | The American University in Cairo | September 2019 – July 2021 | Egypt

- Developed an embedded system integrating four subsystems for sustainable agriculture, enabling real-time agricultural data extraction and analysis using Python, NumPy/Pandas, BeautifulSoup, and VBA.
- Built and evaluated reinforcement learning models for dual-criticality scheduling using OpenAI Gym, RLlib, and TensorFlow, with supporting analysis in NumPy/Pandas and visualization in Matplotlib.

INDUSTRY EXPERIENCE

Software Engineer Co-op | Dell Technologies | March 2022 – March 2023

- Developed production-grade C++ modules for Dell's 5G platform, improving throughput and reliability in latency-sensitive environments.
- Collaborated with distributed engineering teams across Egypt, India, and Canada to design and optimize server deployments, CI/CD pipelines, and gRPC-based microservices.
- Containerized 30+ services using Docker and Podman, reducing deployment time by approximately 15%.

Software Engineer Intern | Silicon Arena LLC | August 2021 – October 2021 | Cairo, Egypt

- Developed front-end modules for an open-source Agile-based e-commerce platform using React.js and REST APIs.
- Collaborated with back-end developers to integrate databases with data-driven user interfaces.

PUBLICATIONS

Conference Papers

A. Potnis, **Y. Hussein**, W. Abebe, J. Lee, D. Varshney, J. Arndt, P. A. Dias, A. Tsaris, D. Lu, D. Lunga. "MOSAIC-CONUS: A Multimodal, Multi-Temporally Paired Dataset for Earth Sciences." *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. Acceptance rate: 25.8%.

- Y. Hussein**, M. F. Mokbel. "KAFY: An Extensible and Scalable Transformers-Based System for Trajectory Data Analysis." *Proceedings of the Very Large Data Bases (PVLDB)*, 19(11), 3385–3399, 2026.
- M. Hemdan, **Y. Hussein**, M. F. Mokbel. "Large Language Models for Spatial Analysis Queries." *2026 IEEE 42nd International Conference on Data Engineering (ICDE)*, pp. 4108–4113, 2026.
- M. F. Mokbel, **Y. Hussein**, M. Hemdan. "Large Language Models for Spatial Analysis Queries." *Proceedings of the Brazilian Symposium on Databases (SBBDB)*, pp. 342–345, 2026.
- Y. Hussein**, M. Hemdan, M. F. Mokbel. "Large Language Models for Spatial Queries Analysis: A Tutorial." *Proceedings of the VLDB Conference*, 2025.
- Y. Huang, A. Uribe, **Y. Hussein**, K. Eldahshoury, G. Ogren, M. F. Mokbel. "POLARIS: An Interactive and Scalable Data Infrastructure for Polar Science." *Proceedings of the VLDB Conference*, 2025.

Demonstrations

- Y. Hussein**, A. Potnis, J. Lee, M. F. Mokbel, D. Lunga. "A Demonstration of Aether: A System for Agentic Retrieval and Reasoning over Multimodal Earth Observation Data." *Proceedings of the ACM SIGSPATIAL*, 2026.
- Y. Hussein**, M. F. Mokbel. "A Demonstration of KAFY: An Extensible and Scalable Transformers-Based System for Trajectory Data Analysis." *Proceedings of the VLDB Endowment (PVLDB)*, 2026.
- Y. Huang, A. Uribe, **Y. Hussein**, K. Eldahshoury, G. Ogren, M. F. Mokbel. "A Demo of POLARIS: An Interactive and Scalable Data Infrastructure for Polar Science." *Proceedings of the VLDB Endowment (PVLDB)*, 2025.

Workshops, Challenges, and Competitions

- Y. Hussein**, J. Lee, O. B. Hoque, D. Lunga. "ARFA: An Agentic System for Real-Time Riverine Flood Response." *ACM SIGSPATIAL 2026 OASIS Student Challenge*, 2026. Top 10 Finalist.
- M. Hemdan, **Y. Hussein**, M. F. Mokbel. "Large Language Models for Spatial Analysis Tasks." *1st ACM SIGSPATIAL International Workshop on Urban Mobility Foundation Models (UMFM)*, 2025.

Journal and Other Publications

- O. A. Hosny Sherif, ..., **Y. Hussein**. "Economic Land Utilization Optimization Model." *Sustainability*, 15(3), 2594, 2023.
- N. Sakr, **Y. Hussein**, K. Farid. "Dual-criticality Scheduling on Non-preemptive, Dynamic Processors Using RL Agents." *The Third International Workshop on Dynamic Scheduling Problems*, pp. 57–62, 2021.

AWARDS & HONORS

- Google DeepMind Student Travel Grant, ACM SIGSPATIAL OASIS — 2026
- OASIS 2026 Top 10 Finalist, ACM SIGSPATIAL Student Challenge — 2026
- VLDB Travel Award Grant, 52nd International Conference on Very Large Databases — 2025
- GAGE Fellowship, Grants to Advance Graduate Education, University of Minnesota — 2023–2024
- Al-Ghurair Foundation Full Scholarship — 2019–2023
- Top Students Dean's Honorary List, The American University in Cairo — 2021–2023
- 2nd Place, Economic Association Data Analysis Competition
- 1st Place, Single-Use Plastic Competition — 2018
- Regional Qualifier, Shell "Imagine The Future" Competition

TEACHING EXPERIENCE

Graduate Teaching Assistant | University of Minnesota, Twin Cities | 2026 | Minneapolis, MN

- CSCI 4707: Practice of Database Systems.

Graduate Teaching Assistant | University of Minnesota, Twin Cities | February 2025 – May 2025 | Minneapolis, MN

- CSCI 5708: Architecture and Implementation of Database Management Systems.
- Supported approximately 50 students through office hours and project guidance, including work involving PostgreSQL.

Undergraduate Teaching Assistant | The American University in Cairo | January 2023 – May 2023 | Cairo, Egypt

- CSCE 4301: Embedded Systems Lab.

Programming Fundamentals Mentor | Udacity | September 2022 – May 2024

- Mentored 300+ students in Python, data structures, and algorithms.
- Delivered 900+ hours of mentoring and technical instruction to students ages 12–17.

Machine Learning Instructor | Google Developer Student Club, AUC | August 2021 – December 2021

- Introduced 40+ students to machine learning concepts, Google BigQuery, and Qwiklabs hands-on exercises.

ACADEMIC SERVICE

Program Committee and Reviewer

- ACM SIGSPATIAL Application Track — 2026
- ACM SIGSPATIAL GeoAI Workshop — 2026

Peer Review

- IEEE Geoscience and Remote Sensing Letters (GRSL) — 2026
- Communications in AI and Computing, Springer — 2026

Conference Organization

- Webmaster, ACM SIGSPATIAL 2026
- Webmaster, ACM SIGSPATIAL 2025

LEADERSHIP & COMMUNITY SERVICE

Board of Directors [Secretary] | Como Student Community Cooperative (CSCC) | August 2025 – May 2026 | Minneapolis, MN

- Volunteer board member for the second consecutive year where I served as the Board Secretary.

Board of Directors [Building Coordinators Chair] | Como Student Community Cooperative (CSCC) | July 2024 – May 2025 | Minneapolis, MN

- Served as liaison between residents and cooperative management through the Building Coordinators Committee.
- Represented residents across 13 buildings and 650+ residents.

Founder & Organizer | Dell Learning Club | April 2022 – Present

- Coordinated a four-person team delivering monthly professional-development sessions across the EMEA region, averaging approximately 50 participants across 8+ countries.

Development Committee Member | AUC Student Union | September 2019 – November 2019

- Developed the Student Union website and integrated the Twitter API using JavaScript, Node.js, and Angular.js.

TECHNICAL SKILLS

Programming: Python, C++, VBA, SQL, Shell Scripting

Machine Learning & AI: PyTorch, TensorFlow, Hugging Face Transformers, Reinforcement Learning, RLlib, OpenAI Gym

Spatial & Scientific Computing: GeoPandas, H3, NumPy, Pandas

Data & Systems: PostgreSQL, MySQL, Docker, Podman, gRPC, CI/CD

Web & Software Development: React.js, Django, JavaScript, REST APIs

Cloud & Development: Google Cloud Platform, Git

REFERENCES

Available upon request.